

Quad 303 Changing the Sensitivity



This is done by changing the gain of the 303. To achieve this, we change the feedback of the input stage. Because we want to lower the sensitivity, we increase the feedback. We change the value of R108 and C103, in such a way that the product of the two values stays constant. This is done to preserve the bandwidth of the 303.

Theoretical values

Sensitivity (V)	R108 (K)	C103 (pF)
0.5 (Standard)	22	100
0.775 (0dBm)	14,19	155
1.0	11	200
1.5	7.33	300

Practical values

Sensitivity (V)	R108 (K)	C103 (pF)
0.5 (Standard)	22	100
0.733	15	150
1.1	10	220
1.6	6.8	330

If you want 1.0V sensitivity just solder the current value of R108 and C103 at the copper side of the pcb, check if the components are not in contact with the bottom chassis plate.

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Color coding of resistors.



To distinguish left from right there is a larger gap between the D and E bands.

- band **A** is the first significant figure of component value (left side)
- band **B** is the second significant figure
- band **C** is the third significant figure
- band **D** is the decimal multiplier
- band **E** indicates tolerance of value in percent

Color	A First figure	B Second figure	C Third figure	D Multiplier		E Tolerance
Black	0	0	0	×1		—
Brown	1	1	1	×10		±1%
Red	2	2	2	×100		±2%
Orange	3	3	3	×1K		—
Yellow	4	4	4	×10K		—
Green	5	5	5	×100K		±0.5%
Blue	6	6	6	×1M		±0.25%
Violet	7	7	7	×10M		±0.1%
Gray	8	8	8	×100M		±0.05%
White	9	9	9	×1G		—
Gold	—	—	—	×0.1		±5%
Silver	—	—	—	×0.01		±10%
None	—	—	—	—		±20%

Example: Red, Red, Black, Red, Brown

220 X 100 = 22Kohm and 1% tolerance

The identification of the plus and minus of electrolyte capacitors.

In almost all cases, the - (minus) is indicated with a long stripe with symbols at the side of the can in the color of the printed text.



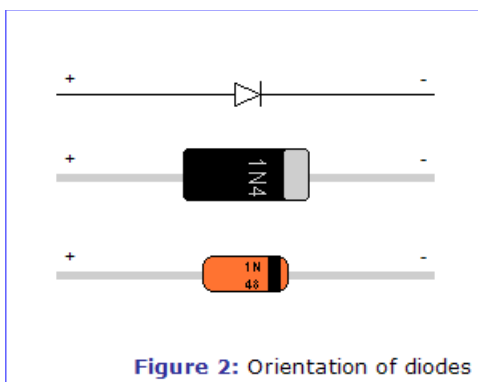
Also if the capacitor has wires, the minus wire is the shortest one!

Capacitors with screw terminals will sometimes have the stripe indication or have indications on *top* of the capacitor, if any doubts, contact us! Be aware: connecting capacitors wrongly could cause serious damage or injuries!



With axial capacitors there is an extra arrow indicating the minus wire, or there is a printed small ring around the body indicating the minus wire. Also the minus wire is direct connected to the aluminum body. The plus wire is sticking through the black plastic cap.

Indication of the cathode of diodes and zener diodes



The cathode will be indicated by a white, silver or black line on the body of the diode.